Notes: Manova (RES, 2013)
Credit Constraints, Heterogeneous Firms, and International Trade

Contents

1 Big picture

- **Question.** How do financial-market imperfections shape countries' export patterns, especially across sectors that differ in external-finance dependence and asset tangibility?
- **Core mechanism.** Heterogeneous firms face liquidity constraints when they must pay fixed and variable costs of exporting before realizing revenue. Better financial development relaxes these constraints and allows more firms to export, more products to be shipped, more destinations to be served, and larger firm-level exports.
- **Key prediction.** Financially developed countries should export relatively more in financially vulnerable sectors: industries that rely heavily on external finance and have few collateralizable tangible assets.
- **Model contribution.** The paper embeds credit constraints in a heterogeneous-firm trade model. Financial frictions operate through both fixed export costs and variable trade costs, so they affect both the extensive margin of exporting and firm-level export intensity.
- **Empirical setting.** The paper uses bilateral exports across 107 exporters, many importers, 27 manufacturing sectors, and years 1985-1995. Financial development is measured mainly by private credit/GDP and alternatively by contract enforcement, accounting standards, and expropriation risk.
- **Main empirical result.** Financially developed countries export more, are more likely to export, export more products, and serve more destinations in sectors with high external finance dependence and low asset tangibility.
- **Decomposition result.** Only about 20-25 percent of the trade effect of credit constraints comes through total production/output. The bulk is trade-specific.
- **Margin result.** Credit constraints affect both firm selection into exporting and average firm-level exports. The intensive margin accounts for a meaningful share but not all of the aggregate trade effect.
- **Economic takeaway.** Financial development is a source of comparative advantage. Countries with deeper financial markets disproportionately specialize in industries that are harder to finance.

2 Incremental contribution of this paper

- **Combines financial frictions with heterogeneous-firm trade.** Earlier work linked finance to growth or aggregate trade; Manova provides a firm-level trade model in which liquidity constraints shape selection, product scope, destination scope, and export volume.
- **Separates output effects from trade-specific effects.** The paper distinguishes the effect of finance on a country's ability to produce from the additional effect of finance on the ability to export. This is a major empirical contribution because it shows that finance matters for trade above and beyond production capacity.
- **Uses sectoral financial vulnerability as a source of identification.** The paper adapts the Rajan-Zingales logic to trade: if finance matters, financially developed countries should have a comparative advantage in sectors that need more outside finance and have less collateral.
- **Identifies multiple export margins.** The paper does not stop at total trade flows. It decomposes the effect into firm selection into exporting, export product variety, number of destinations, and average firm-level exports.
- **Connects institutional measures to trade outcomes.** Beyond private credit/GDP, the paper uses repudiation of contracts, accounting standards, and expropriation risk, showing that broader financial-contracting environments matter.
- **Links theory and reduced-form empirical design tightly.** The interaction terms in the regressions map directly to model predictions about external finance dependence and asset tangibility.

- **Economic significance and counterfactuals.** Tables 7 and 8 quantify how much financial development explains differences and changes in trade, making the paper more than a sign test.
- **Policy implication.** The results imply that domestic financial reforms can reshape comparative advantage and export diversification, especially for financially vulnerable industries.

3 Data and measurement

3.1 Trade outcomes

- Bilateral exports are measured at the exporter-importer-sector-year level and used in logs.
- Firm selection into exporting is proxied by whether an exporter ships positive exports to an importer in a sector-year.
- Product variety is measured by the number of SITC-4 products shipped within an ISIC-3 sector, and robustness uses HS-10 products for exports to the United States.
- Trade partner intensity is the number of destinations served by an exporter in a sector.
- Average firm-level exports are inferred using a selection correction because cross-country firm-level export data are unavailable.

Interpretation: These outcomes map directly to model margins: the probability of exporting, the number of products, the number of destinations, and the conditional value of firm exports.

3.2 Financial development

- The main country-level financial development variable is private credit as a share of GDP.
- Robustness measures include repudiation of contracts, accounting standards, and risk of expropriation.
- These variables capture different aspects of financial contracting: bank-credit depth, enforceability, transparency, and creditor/investor protection.

Interpretation: If the results hold across these measures, the mechanism is less likely to be driven by a single financial-development proxy.

3.3 Sectoral financial vulnerability

- External finance dependence measures how much an industry needs outside financing for investment and working capital.
- Asset tangibility measures how much of a sector's assets can serve as collateral.
- Industries with high external finance dependence and low asset tangibility are more vulnerable to credit constraints.

Interpretation: The key empirical variation is the interaction between country finance and sector vulnerability. This is the paper's main difference-in-differences logic.

3.4 Controls and fixed effects

- Gravity controls include exporter GDP, importer GDP, bilateral distance, and importer CPI.
- Factor-endowment controls interact physical capital, human capital, and natural resources with sector factor intensities.
- Additional institutional controls interact rule of law and corruption with sector financial-vulnerability measures.
- Fixed effects include exporter, importer, sector, year, and in some specifications importer-sector fixed effects.

Interpretation: The design aims to isolate finance-based comparative advantage from ordinary factor-endowment, institutional, market-size, and gravity channels.

4 Models and empirical specifications

4.1 Heterogeneous firms with liquidity constraints

Firms differ in productivity. To export to a destination, a firm must pay fixed costs and variable trade costs before receiving export revenue. If financial markets are imperfect, the firm can borrow only a limited amount based on the country's financial development and the sector's collateral capacity.

Liquidity constraint binds more when: low financial development, high external finance dependence, low asset tangibility.

Interpretation: The model extends Melitz-style selection by adding a financing wedge. Financial development lowers the productivity cutoff for exporting in financially vulnerable sectors.

4.2 Baseline trade-flow regression

$$\log X_{ijst} = \alpha_i + \alpha_j + \alpha_s + \alpha_t + \beta_1 FD_{jt} \times ExtFin_s + \beta_2 FD_{jt} \times Tang_s + \Gamma Z_{ijst} + \varepsilon_{ijst}.$$

- i denotes importer, j exporter, s sector, and t year.
- FD is financial development.
- $ExtFin$ is external finance dependence.
- $Tang$ is asset tangibility.

Interpretation: The model predicts $\beta_1 > 0$ and $\beta_2 < 0$. Financial development should matter more in sectors that need outside finance and have fewer pledgeable assets.

4.3 Extensive-margin regression

$$T_{ijst} = 1\{X_{ijst} > 0\}.$$

The paper estimates Probit models for whether bilateral exports are positive in an exporter-importer-sector-year cell.

Interpretation: This tests whether financial development increases the probability that firms in a country-sector can enter a destination market.

4.4 Product-variety and trade-partner regressions

$$\log Varieties_{ijst} = f(FD_{jt} \times ExtFin_s, FD_{jt} \times Tang_s, controls),$$
$$Partners_{jst} = f(FD_{jt} \times ExtFin_s, FD_{jt} \times Tang_s, controls).$$

Interpretation: These regressions test whether finance expands the scope of trade, not just its value.

4.5 Firm-level export decomposition

The paper uses a two-step correction for firm selection into exporting and then estimates the effect of finance on average firm-level exports:

$$\log X_{ijst} \text{ conditional on exporting} = f(FD_{jt} \times ExtFin_s, FD_{jt} \times Tang_s, selectioncontrols).$$

Interpretation: This decomposes the aggregate trade effect into extensive-margin entry and intensive-margin firm sales.

5 Identification and empirical design

- **Main identification.** Compare financially developed and financially underdeveloped countries across sectors with different financial vulnerability.
- **Key assumption.** Sectoral external finance dependence and tangibility reflect technological needs that are similar across countries and are not chosen in response to each country's financial development.
- **Fixed effects.** Exporter, importer, sector, and year fixed effects remove broad cross-country, market, sectoral, and time variation.
- **Gravity and factor controls.** The specifications control for market size, distance, and interactions between factor endowments and sector factor intensities.
- **Robustness to other institutions.** The paper adds rule of law, corruption, GDP per capita, and their interactions to reduce the concern that finance proxies general institutional quality.
- **Margin consistency.** The same interaction signs appear across total exports, positive export probability, product variety, trade partners, and firm-level exports.
- **Limitations.** The paper is not based on a randomized financial reform; identification relies on cross-country-sector interactions and the plausibility of sectoral financial vulnerability measures.

6 Section-by-section study guide

- **Introduction.** Establishes the puzzle: financially developed countries export disproportionately in sectors that are hard to finance.
- **Model.** Adds credit constraints to heterogeneous-firm trade and derives predictions for export value, extensive margin, product variety, and destinations.
- **Data.** Defines financial development, external finance dependence, asset tangibility, bilateral exports, product variety, and trade partners.
- **Trade versus production.** Tables 1-2 show that finance affects trade beyond overall production.
- **Export margins.** Tables 3-6 decompose the effect into selection, product variety, destinations, and firm-level exports.

- **Economic significance.** Tables 7-8 quantify the size of the effect and the predictive power of financial development.
- **Appendix.** Appendix tables define financial development, sector characteristics, export patterns, and the decomposition of firm selection versus firm exports.

7 Tables and figures

7.1 Figure 1: The effects of financial constraints on trade

Read:

- Panel A shows export profits when liquidity constraints affect fixed costs only.
- Panel B shows export profits when liquidity constraints affect both fixed and variable costs.
- Credit constraints raise export productivity cutoffs and reduce export profits for constrained firms.

Detailed interpretation: Panel B is especially important because the paper's empirical decomposition relies on the idea that finance affects both fixed-cost entry and variable-cost export intensity. If constraints only affected fixed costs, finance would mostly shape the extensive margin. If constraints also affect variable costs, conditional firm exports also fall.

Economic message: Financial underdevelopment raises the productivity threshold for exporting and reduces the scale of exporting firms that do enter.

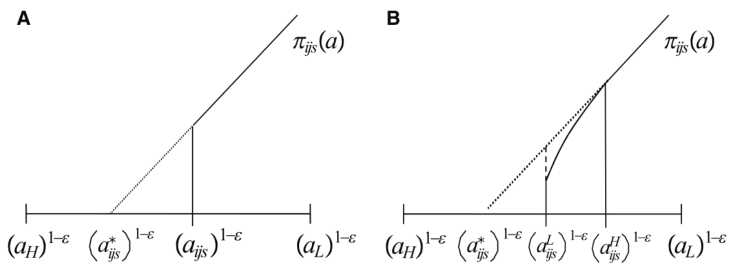


FIGURE 1

The effects of financial constraints on trade. This figure plots export profits as a function of productivity. It shows th wedge between the productivity cut-offs for exporting with and without credit constraints in the financing of fixed co only (A) and of both fixed and variable costs (B). Panel B also shows the lower profits earned by firms with productiv below the cut-off for exporting at first-best levels

Figure 1: The effects of financial constraints on trade.

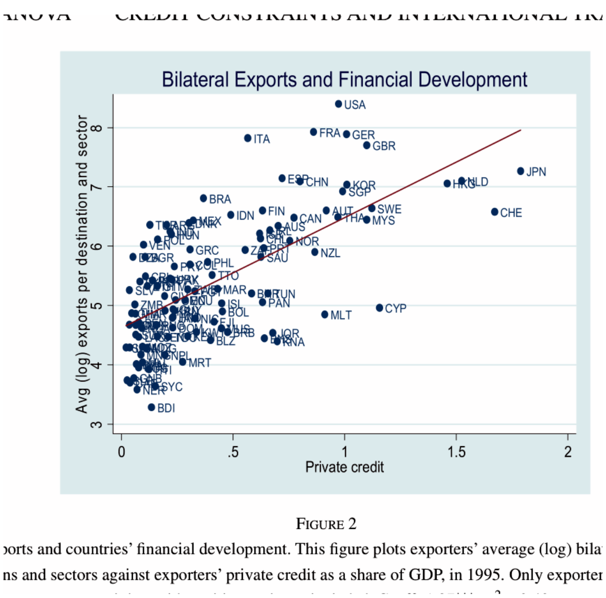
7.2 Figures 2-3: Reduced-form patterns in the raw data

Read:

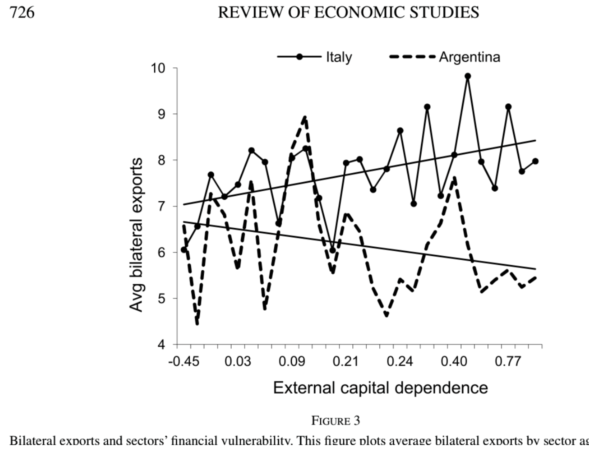
- Figure 2 plots average bilateral exports against private credit and shows a strong positive relationship.
- Figure 3 compares Italy and Argentina across sectors ordered by external finance dependence.
- Italy's export advantage over Argentina is larger in more financially vulnerable sectors.

Detailed interpretation: These figures are not the causal identification, but they motivate the regressions. Figure 2 shows the country-level relationship between finance and trade. Figure 3 shows the interaction logic: the finance advantage is larger where financing needs are higher.

Economic message: The raw patterns already suggest finance-based comparative advantage.



(a) Figure 2: bilateral exports and financial development.



(b) Figure 3: exports and sector financial vulnerability.

7.3 Tables 1-2: Financial constraints and trade versus production

Read:

- Table 1 shows that financial development interacts positively with external finance dependence and negatively with asset tangibility in export regressions.
- The signs are stable after controlling for domestic production, selection into production, CPI interactions, and importer-sector fixed effects.
- Table 2 shows robustness across alternative financial-development measures and factor/institution controls.

Detailed interpretation: These are the paper's central reduced-form tables. They establish that finance matters for exports beyond its effect on domestic output. The positive coefficient on $FD \times ExtFin$ and negative coefficient on $FD \times Tang$ mean that financial development disproportionately boosts exports in sectors where financing is harder.

Economic message: Financial markets create comparative advantage, and the effect is not just a by-product of producing more at home.

TABLE 1
Financial constraints and trade vs. production

Financial development measure: Private credit
Dependent variable: m_{ijst}, (log) bilateral exports by sector

	Total effect of credit constraints	Controlling for selection into domestic production		Proxy for p_{is}: CPI and interactions with sector FE	Proxy for p_{is}: Importer's consumption by sector	Proxy for p_{is}: Importer sector FI
Fin devt	0.167	0.251	0.022	0.225	0.267	0.306
	(3.14)***	(4.25)***	(0.37)	(3.64)***	(4.54)***	(5.26)*
Fin devt × Ext fin dep	1.752	1.296	1.489	1.343	1.253	1.372
	(43.29)***	(28.31)***	(30.47)***	(29.01)***	(26.36)***	(33.87)*
Fin devt × Tang	−2.624	−2.130	−2.077	−2.204	−2.171	−2.434
	(−24.65)***	(−16.41)***	(−17.75)***	(−16.64)***	(−16.45)***	(−19.46)*
(Log) # Establish		0.318		0.321	0.323	0.321
		(40.47)***		(39.89)***	(40.66)***	(42.34)*
(Log) Output			0.316			
			(18.52)***			
p_{is}				0.008	0.169	
				(6.86)***	(26.74)***	
LGDPE	0.957	1.079	0.667	1.071	1.082	1.119
	(16.75)***	(16.17)***	(9.38)***	(16.05)***	(16.29)***	(16.64)*
LGDPI	0.949	0.980	0.946	1.040	0.711	0.998
	(16.55)***	(14.41)***	(14.49)***	(16.36)***	(10.28)***	(14.57)*
LDIST	−1.374	−1.408	−1.410	−1.418	−1.414	−1.442
	(−79.05)***	(−72.20)***	(−74.24)***	(−70.27)***	(−71.74)***	(−73.35)*
Controls:						
Exporter, Year FE	Y	Y	Y	Y	Y	Y
Importer, Sector FE	Y	Y	Y	Y	Y	N
Importer × Sector FE	N	N	N	N	N	Y
R^2	0.57	0.57	0.59	0.58	0.58	0.60
# observations	861,380	621,333	703,743	579,485	589,205	621,333
# exporter–importer clusters	9343	7867	8031	7452	7813	7867
# exporters	107	95	94	95	95	95

Notes: This table examines the effect of credit constraints on trade flows above and beyond that on overall productio The dependent variable is (log) exports from country *j* to country *i* in a 3-digit ISIC sector *s* and year *t*, 1985–199 Financial development is measured by private credit. External finance dependence *Ext fin dep* and asset tangibility *Tai* are defined in the text. *(Log) # Establish* and *(Log) Output* are the (log) number of domestic establishments and (lo output in the exporting country by year and sector. The sectoral price index in the importing country is proxied by tl importer's consumer price index (CPI) and its interactions with sector dummies in Column 4; the importer's consumptic

(a) Table 1: trade versus production.

TABLE 2
Financial constraints and trade vs. production: robustness

Dependent variable: m_{ijst}, (log) bilateral exports by sector

Financial development measure:		Private credit		Repudiation of contracts	Accounting standards	Risk of expropriati
Fin devt	−0.439	0.743	−0.019			
	(−8.62)***	(11.64)***	(−0.24)			
Fin devt × Ext fin dep	1.408		1.101	0.576	0.025	0.551
	(30.06)***		(15.38)***	(19.34)***	(11.46)***	(14.38)**
Fin devt × Tang		−2.472	−1.334	−1.488	−0.071	−1.474
		(−18.37)***	(−6.64)***	(−15.78)***	(−11.12)***	(−12.58)**
(Log) # Establish	0.321***	0.360***	0.314***	0.302***	0.306***	0.305*
Importer's CPI	0.008***	0.008***	0.008***	0.008***	0.009***	0.008*
Physical capital per Worker, K/L			0.420***	0.375***	0.042	0.364*
Human capital per Worker, H/L			−1.350***	−1.323***	−1.003***	−1.308*
Natural resources per Worker, N/L			1.357***	1.533***	2.721***	1.577*
K/L × Industry K intensity			−1.491***	−1.470***	−0.848*	−1.362*
H/L × Industry H intensity			1.435***	1.398***	1.225***	1.385*
N/L × Industry N intensity			0.219***	0.207***	0.282***	0.204*
LGDPCE			−2.984***	−3.453***	−5.531***	−3.379*
LGDPCE × Ext fin dep			0.453***	0.054	0.491***	0.390*
LGDPCE × Tang			−0.471**	0.804***	−0.433*	0.024
Rule of law × Ext fin dep			0.060***	−0.041*	0.131***	−0.097*
Rule of law × Tang			0.244***	0.537***	−0.182**	0.673*
Corruption × Ext fin dep			−0.193***	−0.185***	−0.224***	−0.182*
Corruption × Tang			−0.139**	−0.083	0.294***	−0.089
Controls:	LGDPE, LGDPI, LDIST, CPI × Sector FE, Exporter, Importer, Year, and Sector FE					
R^2	0.58	0.58	0.59	0.59	0.61	0.59
# observations	579,485	579,485	428,444	436,931	396,112	436,931
# exporter-importer clusters	7452	7452	4130	4132	3374	4132
# exporters	95	95	40	40	32	40

Notes: This table examines the robustness of the effect of credit constraints on trade flows above and beyond that overall production. The dependent variable is the (log) value of exports from country *j* to country *i* in a 3-digit ISIC sec *s* and year *t*, 1985–1995. The measure of financial development is indicated by the column heading. External finar dependence *Ext fin dep* and asset tangibility *Tang* are defined in the text. All regressions control for the exporter's (lc number of domestic establishments; the importer's CPI and its interactions with sector dummies; the (log) real GDP both trade partners and the (log) distance between them. Columns 3–6 also control for the exporter's factor endowme (natural resources, physical, and human capital) and their interactions with sector factor intensities; the exporter's GDP I capita *LGDPCE*; and the interactions of *LGDPCE*, rule of law and corruption with *Ext fin dep* and *Tang*. All regressic include a constant term, exporter, importer, sector, and year fixed effects, and cluster errors by exporter–importer pa *T*-statistics in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level.

(b) Table 2: robustness.

7.4 Tables 3-4: Firm selection and export product variety

Read:

- Table 3 shows that financial development increases the probability of positive bilateral exports more in financially vulnerable sectors.
- Table 4 shows that financial development raises the number of products exported bilaterally, especially in sectors with high external finance dependence and low tangibility.
- Both tables use alternative financial-development measures and extensive controls.

Detailed interpretation: These tables move from trade values to export margins. Table 3 establishes that finance affects market entry: whether any firms export to a market. Table 4 establishes that finance affects scope: how many product varieties are shipped.

Economic message: Credit constraints reduce international trade by keeping firms and products out of foreign markets.

TABLE 3

Financial constraints and firm selection into exporting

Dependent variable: T_{ijst}, indicator variable equal to 1 when positive bilateral exports in a sector

Financial development measure:	Private credit	Repudiation of contracts	Accounting standards	Risk of expropriatio
Fin devt	−0.110 (−2.09)**			
Fin devt × Ext fin dep	1.029 (19.86)***	0.320 (19.51)***	0.022 (17.46)***	0.435 (21.06)***
Fin devt × Tang	−0.823 (−8.23)***	−0.537 (−14.00)***	−0.028 (−8.79)***	−0.522 (−11.08)***
Importer's CPI	0.007***	0.007***	0.007***	0.007***
LGDPE	4.682***	4.972***	7.388***	4.966***
LGDPI	0.369***	0.382***	0.403***	0.383***
LDIST	−1.076***	−1.086***	−1.161	−1.087***
(Log) # Procedures	−0.719***	−0.726***	−0.763***	−0.755***
(Log) # Days	0.057	0.047	−0.057	0.052
(Log) Cost	−0.207***	−0.214***	−0.153***	−0.209***
Controls:	LGDPE, LGDPI, LDIST, Exp, Imp, Year, Sector FE, CPI × Sector FE K, H, N, LGDPCE, Institutions, and Interactions			
Pseudo R^2	0.51	0.51	0.51	0.51
# observations	1,079,865	1,103,274	906,390	1,103,274
# exporter–importer clusters	3965	3965	3259	3965

Notes: This table examines the effect of credit constraints on firm selection into exporting. The dependent variable is indicator variable equal to 1 if country j exports to country i in a 3-digit ISIC sector s and year t, 1985–1995. The measu of financial development is indicated by the column heading. External finance dependence *Ext fin dep* and asset tangibil *Tang* are defined in the text. All regressions control for the average number of procedures and days it takes to establis business in the exporting and importing countries, and the cost of doing so as a share of GDP per capita. All regressic include a constant term, exporter, importer, sector, and year fixed effects; the importer's CPI and its interactions w sector dummies; the (log) real GDP of both partners and the (log) distance between them; factor endowments, institutio GDP per capita, and their interactions as in Table 2. Errors clustered by exporter–importer pair. T-statistics in parenthes ***, **, and * indicate significance at the 1%, 5%, and 10% level.

(25th percentile). The corresponding differential effect across sectors at different levels of ass tangibility is 17%. The estimated impact of a one-standard-deviation improvement in priva credit is somewhat smaller. These effects are on par with those of a one-standard-deviation ri

(a) Table 3: firm selection into exporting.

TABLE 4

Financial constraints and export product variety

Financial development measure:	Private credit		Repudiation of contracts	Accounting standards	Risk of expropriation
Panel A. Dep variable: x_{ijst}, (log) # SITC-4 products exported bilaterally by sector					
Fin devt	−0.086 (−3.83)***	−0.089 (−3.17)***			
Fin devt × Ext fin dep	0.405 (28.67)***	0.335 (16.37)***	0.176 (18.45)***	0.008 (11.74)***	0.190 (16.32)***
Fin devt × Tang	−0.455 (−10.46)***	−0.400 (−6.07)***	−0.272 (−10.10)***	−0.014 (−7.14)***	−0.268 (−8.00)***
(Log) # Establish	0.098***	0.092***	0.090***	0.091***	0.091***
Importer's CPI	0.007***	0.008***	0.008***	0.009***	0.008***
Controls:	LGDPE, LGDPI, LDIST, Exp, Imp, Year, Sector FE, CPI × Sector FE	K, H, L, LGDPCE, Institutions, and Interactions			
R^2	0.63	0.64	0.64	0.65	0.64
# observations	579,485	428,444	436,931	396,112	436,931
# exporter–importer clusters	7452	4130	4132	3374	4132
# exporters	95	40	40	32	40
Panel B. Dep variable: x_{ijst}, (log) # HS-10 products exported to the U.S. by sector					
Fin devt	−0.111 (−0.78)	0.332 (1.47)			
Fin devt × Ext fin dep	0.802 (5.07)***	0.518 (2.74)***	0.346 (5.13)***	0.020 (3.68)***	0.326 (3.05)***
Fin devt × Tang	0.360 (1.08)	−0.148 (−0.36)	−0.293 (−1.31)	−0.034 (−2.15)**	−0.242 (−0.79)
(Log) # Establish	0.213***	0.185***	0.179***	0.189***	0.183***
Controls:	LGDPE, Exporter, Year and Sector FE, CPI × Sector FE	K, H, N, LGDPCE, Institutions, and Interactions			
R^2	0.86	0.89	0.89	0.90	0.89
# observations	9605	5836	5916	4899	5916
# exporters	87	38	38	30	38

Notes: This table examines the effect of credit constraints on export product variety. The dependent variable in Panel A the (log) number of 4-digit SITC products country j exports to country i in a 3-digit ISIC sector s and year t, 1985–199 The dependent variable in Panel B is the (log) number of 10-digit HS products j exports to the U.S. in a 3-digit ISI sector s and year t, 1989–1995. The measure of financial development is indicated by the column heading. Extern finance dependence *Ext fin dep* and asset tangibility *Tang* are defined in the text. All regressions include a constant tern exporter, importer, sector, and year fixed effects; the exporter's (log) number of domestic establishments; the importer CPI and its interactions with sector dummies; the (log) real GDP of both partners and the (log) distance between then and cluster errors by exporter–importer pair. In Panel B, bilateral distance, importer GDP, CPI, and importer fixed effec are dropped, and errors clustered by exporter. Columns 2–5 control for factor endowments, institutions, GDP per capit

(b) Table 4: export product variety.

7.5 Tables 5-6: Trade partners and firm-level exports

Read:

- Table 5 shows that financial development increases the number of export destinations, particularly in financially vulnerable sectors.
- Table 6 estimates average firm-level exports after correcting for firm selection into exporting.
- Table 6 shows that credit constraints also reduce conditional firm export volumes, not only entry.

Detailed interpretation: Table 5 captures geographical scope. Table 6 captures intensive margin conditional on exporting. Together they show that finance affects how many markets a country serves and how much surviving exporters sell.

Economic message: Financial frictions matter along multiple margins: entry, product scope, destination scope, and firm-level export scale.

TABLE 5
Financial constraints and trade partner intensity

Dependent variable: I_{js}, number of trade partners by sector					
Financial development measure:	Private credit		Repudiation of contracts	Accounting standards	Risk of expropriation
Panel A. Whole sample					
Fin devt	−10.61	−4.71			
	(−2.29)**	(−0.71)			
Fin devt × Ext fin dep	51.73	28.40	11.29	0.68	15.74
	(15.27)***	(4.05)***	(4.79)***	(3.91)***	(6.24)***
Fin devt × Tang	8.20	−12.92	−10.56	−0.65	−10.68
	(1.03)	(−0.87)	(−2.73)***	(−1.97)*	(−1.96)*
LRGDPE	18.09	105.86	111.59	218.66	111.29
	(3.79)***	(2.53)**	(2.63)**	(5.41)**	(2.63)**
Controls:	Exporter, Year and Sector Fixed Effects; K, H, N, LGDPCE, Institutions, and Interactions				
R^2	0.88	0.86	0.87	0.87	0.87
# observations	30,296	12,656	12,936	10,472	12,936
# exporters	107	42	42	34	42
Panel B. Sample with nonzero partners					
Fin devt	−2.23	−0.96			
	(−0.46)	(−0.14)			
Fin devt × Ext fin dep	41.94	24.04	9.57	0.59	12.86
	(13.44)***	(3.66)***	(4.37)***	(3.58)***	(5.40)***
Fin devt × Tang	−17.04	−22.68	−15.11	−0.87	−18.15
	(−2.12)**	(−1.55)	(−3.90)***	(−2.72)***	(−3.44)***
LRGDPE	19.99	111.00	117.36	227.55	117.75
	(3.88)***	(2.56)**	(2.67)**	(5.42)***	(2.68)**
Controls:	Exporter, Year and Sector Fixed Effects; K, H, N, LGDPCE, Institutions, and Interactions				
R^2	0.90	0.87	0.87	0.88	0.87
# observations	26,900	12,170	12,440	10,088	12,440
# exporters	107	42	42	34	42

Notes: This table examines the effect of credit constraints on the number of countries' trading partners. The depende variable is the number of country *j*'s export destinations in a 3-digit ISIC sector *s* and year *t*, 1985–1995. Panel presents results for the full matrix of exporter–sector pairs, whereas Panel B restricts the sample to exporter–sector ye observations with at least 1 trade partner. The measure of financial development is indicated by the column headin External finance dependence *Ext fin dep* and asset tangibility *Tang* are defined in the text. All regressions include constant term, exporter, sector, and year fixed effects, and cluster errors by exporter. Columns 2–5 control for fact endowments, institutions, GDP per capita, and their interactions as in Table 2. *T*-statistics in parentheses. ***, **, and *

(a) Table 5: trade partner intensity.

TABLE 6
Financial constraints and firm-level exports

Dependent variable: m_{ijst}, (log) bilateral exports by sector				
Financial development measure:	Private credit	Repudiation of contracts	Accounting standards	Risk of expropriation
Panel A. Maximum likelihood estimation				
Fin devt	0.028			
	(0.34)			
Fin devt × Ext fin dep	0.409	0.369	0.012	0.277
	(4.07)***	(10.22)***	(4.71)***	(5.80)***
Fin devt × Tang	−0.803	−1.182	−0.052	−1.123
	(−3.72)***	(−11.40)***	(−7.78)***	(−9.05)***
delta (from w_{ijs})	0.806	0.820	0.758	0.817
	(7.91)***	(8.25)***	(8.55)***	(8.24)***
eta_{ijs}	0.909	0.877	0.874	0.875
	(9.63)***	(9.49)***	(10.86)***	(9.55)***
(Log) # Establish	0.305***	0.294***	0.297***	0.297***
Importer's CPI	0.004***	0.004***	0.005***	0.004***
Controls:	LGDPE, LGDPI, LDIST, Exp, Imp, Year, Sector FE, CPI × Sector FE; K, H, N, LGDPCE, Institutions, and Interactions			
# observations	398,726	406,677	367,634	406,677
# exporter–importer clusters	3681	3682	2995	3682
Panel B. More flexible specification: OLS with polynomial in z_{ijs}				
Fin devt	0.030			
	(0.38)			
Fin devt × Ext fin dep	0.357	0.360	0.012	0.250
	(3.75)***	(10.36)***	(4.87)***	(5.40)***
Fin devt × Tang	−0.777	−1.165	−0.052	−1.078
	(−3.63)***	(−11.48)***	(−7.81)***	(−8.79)***
z_{ijs}	3.388	3.346	2.828	3.308
	(15.77)***	(15.68)***	(12.93)***	(15.43)***
$(z_{ijs})^2$	−0.653	−0.635	−0.500	−0.625
	(−9.38)***	(−9.12)***	(−7.00)***	(−8.90)***
$(z_{ijs})^3$	0.049	0.047	0.034	0.046
	(6.35)***	(6.05)***	(4.32)***	(5.88)***
eta_{ijs}	1.479	1.452	1.380	1.438
	(16.66)***	(16.68)***	(16.38)***	(16.43)***
(Log) # Establish	0.306***	0.296***	0.297***	0.298***
Importer's CPI	0.004***	0.004***	0.005***	0.004***
Controls:	LGDPE, LGDPI, LDIST, Exp, Imp, Year, Sector FE, CPI × Sector FE; K, H, N, LGDPCE, Institutions, and Interactions			
R^2	0.62	0.62	0.63	0.62
# observations	398,726	406,677	367,634	406,677
# exporter–importer clusters	3681	3682	2995	3682

(b) Table 6: firm-level exports, panels A-B.

TABLE 6
Continued

Dependent variable: m_{ijst}, (log) bilateral exports by sector				
Financial development measure:	Private credit	Repudiation of contracts	Accounting standards	Risk of expropriatio
Panel C. Most flexible specification: OLS with 50 bins for predicted probability of exporting				
Fin devt	0.010			
	(0.12)			
Fin devt × Ext fin dep	0.491	0.401	0.013	0.303
	(5.79)***	(12.44)***	(5.36)***	(7.08)***
Fin devt × Tang	−0.881	−1.235	−0.054	−1.144
	(−4.17)***	(−12.43)***	(−8.07)***	(−9.44)***
(Log) # Establish	0.306***	0.296***	0.298***	0.299***
Importer's CPI	0.005***	0.004***	0.006***	0.004***
Controls:	LGDPE, LGDPI, LDIST, Exp, Imp, Year, Sector FE, CPI × Sector FE; K, H, N, LGDPCE, Institutions, and Interactions			
R^2	0.62	0.62	0.63	0.62
# observations	398,726	406,677	367,634	406,677
# exporter–importer clusters	3681	3682	2995	3682

Notes: This table examines the effect of credit constraints on average firm-level exports. The dependent variable is (lc

(a) Table 6: firm-level exports, panel C.

at the 1%, 5%, and 10% level.

TABLE 7
Economic significance: comparative statics

One st. dev. increase in:	Private credit		Repudiation of contracts		K Endow	H Endo
Differential effect across sectors at different levels of:	Ext Fin Dep	Asset Tang	Ext Fin Dep	Asset Tang	K Intensity	H Intens
Trade outcome: 1. Total bilateral exports (%)	15	14	20	30	−9	32
2. Probability of bilateral exports (%)	14	6	19	17	3	15
3. Bilateral export product variety (%)	5	3	10	8	3	11
4. Trade partner intensity	3.2	1.6	5.7	4.8	−0.2	4.4
5. (Avg.) bilateral firm exports (%)	6	6	15	25	−4	30

Notes: This table examines the economic significance of the effects of credit constraints on trade. Each cell reports a different comparative static exercise based on coefficient estimates from regressions in Tables 2–6. The relevant tra outcome is indicated in the row heading. All values are in percentage points, except for the change in trade partı intensity which is in absolute levels. Column 1 (Column 3) shows how much bigger the effect of a one-standard-deviati increase in private credit (repudiation of contracts) is on the sector at the 75th percentile of the distribution by exteri finance dependence relative to the sector at the 25th percentile. Column 2 (Column 4) shows how much bigger the eff of a one-standard-deviation increase in private credit (repudiation of contracts) is on the sector at the 25th percentile

(b) Table 7: economic significance.

7.6 Tables 7-8: Economic significance and predicted versus actual trade growth

Read:

- Table 7 translates coefficients into comparative statics from a one-standard-deviation improvement in finance or institutions.
- Table 8 compares the predictive power of financial development and factor accumulation for world export levels and export growth.
- Financial development explains large differences in levels and meaningful variation in changes.

Detailed interpretation: Table 7 makes the coefficient magnitudes interpretable in percentage terms. Table 8 is important because it compares finance to traditional factor accumulation; finance has substantial explanatory power for trade patterns.

Economic message: The effects are economically large, not only statistically significant.

at the 1%, 5%, and 10% level.

TABLE 7
Economic significance: comparative statics

One st. dev. increase in:	Private credit		Repudiation of contracts		K Endow	H Endo
Differential effect across sectors at different levels of:	Ext Fin Dep	Asset Tang	Ext Fin Dep	Asset Tang	K Intensity	H Intens
Trade outcome: 1. Total bilateral exports (%)	15	14	20	30	−9	32
2. Probability of bilateral exports (%)	14	6	19	17	3	15
3. Bilateral export product variety (%)	5	3	10	8	3	11
4. Trade partner intensity	3.2	1.6	5.7	4.8	−0.2	4.4
5. (Avg.) bilateral firm exports (%)	6	6	15	25	−4	30

Notes: This table examines the economic significance of the effects of credit constraints on trade. Each cell reports a different comparative static exercise based on coefficient estimates from regressions in Tables 2–6. The relevant tra outcome is indicated in the row heading. All values are in percentage points, except for the change in trade part intensity which is in absolute levels. Column 1 (Column 3) shows how much bigger the effect of a one-standard-deviati increase in private credit (repudiation of contracts) is on the sector at the 75th percentile of the distribution by exter finance dependence relative to the sector at the 25th percentile. Column 2 (Column 4) shows how much bigger the eff of a one-standard-deviation increase in private credit (repudiation of contracts) is on the sector at the 25th percentile

(a) Table 7: comparative statics.

TABLE 8
Economic significance: predicted vs. actual trade growth

Financial development measure: private credit

Dependent variable:	Actual level of world exports in 1995 (Beta coefficients)			Actual change in world exports (Beta coefficients)		
Predicted Value of Dep Var due to: Financial development	0.92***		1.02***	0.47***		0.40**
Factor accumulation		0.80***	−0.11		0.34***	0.19**
R^2	0.85	0.65	0.85	0.22	0.12	0.25
# observations	4508	4508	4508	4508	4508	4508
# exporters	161	161	161	161	161	161

Notes: This table examines the predictive power of financial development and factor accumulation for explaining chang

(b) Table 8: predicted versus actual trade growth.

7.7 Appendix Tables A1-A4: Measurement and decomposition details

Read:

- Table A1 reports country-level financial-development measures.
- Table A2 reports sectoral external finance dependence, tangibility, and factor intensities.
- Table A3 reports summary statistics for export outcomes.
- Table A4 decomposes the trade-specific effect into firm selection and firm-level export margins.

Detailed interpretation: The appendix tables are essential for replication and interpretation. They define the cross-country and cross-sector variables that drive the interaction design, and Table A4 provides a compact summary of the extensive versus intensive decomposition.

Economic message: The appendix connects the reduced-form regressions to the measurement of finance, sector vulnerability, and export margins.

APPENDIX

TABLE A1

Countries' financial development

Panel A. Private credit

Country	Avg	St Dev	Country	Avg	St Dev	Country	Avg	St Dev
Algeria	0.35	0.22	Germany	0.93	0.04	Nigeria	0.14	0.04
Argentina	0.14	0.03	Ghana	0.04	0.01	Norway	0.87	0.10
Australia	0.54	0.14	Greece	0.37	0.07	Pakistan	0.24	0.02
Austria	0.87	0.06	Guatemala	0.14	0.02	Panama	0.47	0.07
Bangladesh	0.15		Guinea-Bissau[a]	0.03	0.02	Papua New Guinea	0.23	0.05
Barbados	0.42	0.05	Guyana	0.23		Paraguay	0.16	0.05
Belize	0.37	0.03	Haiti	0.11	0.02	Peru	0.09	0.03
Benin	0.11	0.03	Honduras	0.29	0.04	Philippines[b]	0.23	0.08
Bolivia	0.24	0.14	Hong Kong	1.35	0.09	Poland	0.11	0.08
Brazil[c]	0.24	0.08	Hungary	0.33	0.11	Portugal[d]	0.58	0.09
Bulgaria	0.06	0.03	Iceland	0.40	0.06	Rwanda	0.09	0.02
Burkina Faso	0.13	0.03	India	0.26	0.04	Senegal	0.27	0.05
Burundi	0.09	0.03	Indonesia	0.33	0.13	Seychelles	0.10	0.02

(continued

(a) Table A1: financial development, part 1.

TABLE A1

Continued

Panel A. Private credit

Country	Avg	St Dev	Country	Avg	St Dev	Country	Avg	St De
Cameroon	0.20	0.07	Iran	0.29	0.03	Sierra Leone	0.03	0.00
Canada	0.73	0.06	Ireland	0.63	0.02	Singapore	0.95	0.06
Centr Afr Rep	0.07	0.02	Israel	0.53	0.05	South Africa	0.50	0.03
Chad	0.10	0.05	Italy	0.54	0.05	South Korea	0.80	0.13
Chile	0.51	0.07	Jamaica	0.26	0.04	Spain	0.77	0.05
China	0.78	0.04	Japan[e]	1.63	0.16	Sri Lanka	0.16	0.05
Colombia	0.24	0.07	Jordan	0.67	0.05	St Kitts and Nevis	0.54	0.11
Congo	0.12	0.04	Kenya	0.29	0.02	Sweden	1.15	0.17
Costa Rica	0.14	0.03	Madagascar	0.15	0.02	Switzerland	1.55	0.11
Cote d'Ivoire	0.33	0.06	Malawi	0.10	0.02	Syrian Arab Rep	0.08	0.01
Cyprus	0.87	0.23	Malaysia	0.85	0.17	Thailand	0.64	0.18
Denmark	0.43	0.08	Mali	0.12	0.02	Togo	0.24	0.03
Dominican Rep	0.24	0.03	Malta	0.72	0.15	Trinidad & Tobago	0.48	0.05
Ecuador	0.18	0.05	Mauritania	0.33	0.06	Tunisia	0.56	0.07
Egypt	0.29	0.03	Mauritius	0.32	0.07	Turkey	0.14	0.01
El Salvador	0.04	0.02	Mexico	0.19	0.09	Uganda	0.02	0.01
Equator Guinea	0.18	0.07	Morocco	0.25	0.13	United Kingdom	0.95	0.23
Ethiopia	0.16	0.03	Mozambique	0.10	0.01	United States	0.91	0.05
Fiji	0.33	0.06	Nepal	0.12	0.03	Uruguay	0.25	0.05
Finland	0.74	0.13	Netherlands	1.29	0.18	Venezuela	0.31	0.14
France	0.86	0.08	New Zealand	0.63	0.24	Zambia	0.06	0.02
Gabon	0.15	0.06	Nicaragua	0.18	0.13	Zimbabwe	0.20	0.06
Gambia	0.13	0.04	Niger	0.13	0.04			

Average in the cross-section: 0.39 — Average in the panel: 0.40

Standard deviation in the cross-section: 0.34 — Standard deviation in the panel: 0.35

Panel B. Other measures of financial development

Financial Devt Measure	# Obs	Average	St Dev	Min	Ma
Repudiation of contracts	49	7.58	1.79	4.36	9.9
Accounting standards	41	60.93	13.40	24	83
Risk of expropriation	49	8.05	1.59	5.22	9.9

Notes: This table summarizes the variation in financial development in the data. Panel A reports the time-series mean a
standard deviation for each country in the sample, as well as summary statistics for the cross-section of means and
the entire panel, 1985–1995. Panel B presents summary statistics for repudiation of contracts, accounting standards, a
the risk of expropriation, which vary only in the cross-section. [a,b,c,d,e] identify the country with the lowest, 1st quarti
median, 3rd quartile, and highest level of private credit.

TABLE A2

Industry characteristics

ISIC code	Industry	External finance dependence	Asset tangibility	Physical capital intensity	Human capital intensity	Natura resourc intensit
311	Food products	0.1368	0.3777	0.0616	0.8117	0
313	Beverages	0.0772	0.2794	0.0620	1.1345	0
314	Tobacco	−0.4512	0.2208	0.0181	1.3539	0
321	Textiles	0.4005	0.3730	0.0726	0.6881	0
322	Wearing apparel, except footwear	0.0286	0.1317	0.0189	0.5017	0
323	Leather products	−0.1400	0.0906	0.0324	0.6869	0
331	Wood products, except furniture	0.2840	0.3796	0.0653	0.7409	1

(continue

(b) Table A1 part 2 and Table A2 start.

TABLE A2
Continued

ISIC code	Industry	External finance dependence	Asset tangibility	Physical capital intensity	Human capital intensity	Natura resourc intensit
332	Furniture, except metal	0.2357	0.2630	0.0390	0.6984	0
341	Paper and products	0.1756	0.5579	0.1315	1.1392	1
342	Printing and publishing	0.2038	0.3007	0.0515	0.9339	0
352	Other chemicals	0.2187	0.1973	0.0597	1.2089	0
353	Petroleum refineries	0.0420	0.6708	0.1955	1.6558	1
354	Misc. petroleum and coal products	0.3341	0.3038	0.0741	1.1531	1
355	Rubber products	0.2265	0.3790	0.0656	0.9854	0
356	Plastic products	1.1401	0.3448	0.0883	0.8274	0
361	Pottery, china, earthenware	−0.1459	0.0745	0.0546	0.8041	0
362	Glass and products	0.5285	0.3313	0.0899	1.0121	0
369	Other non-metallic products	0.0620	0.4200	0.0684	0.9522	1
371	Iron and steel	0.0871	0.4581	0.1017	1.2510	1
372	Non-ferrous metals	0.0055	0.3832	0.1012	1.0982	1
381	Fabricated metal products	0.2371	0.2812	0.0531	0.9144	0
382	Machinery, except electrical	0.4453	0.1825	0.0582	1.1187	0
383	Machinery, electric	0.7675	0.2133	0.0765	1.0636	0
384	Transport equipment	0.3069	0.2548	0.0714	1.3221	0
385	Prof and scient equipment	0.9610	0.1511	0.0525	1.2341	0
390	Other manufactured products	0.4702	0.1882	0.0393	0.7553	0
3511	Industrial chemicals	0.2050	0.4116	0.1237	1.4080	0
	Industry average	0.2534	0.3044	0.0714	1.0168	0.2593
	Industry standard deviation	0.3301	0.1372	0.0369	0.2666	0.4466

Notes: This table reports the measures of external finance dependence, asset tangibility, and factor intensity with respe to natural resources, physical, and human capital for all 27 3-digit ISIC sectors used in the empirical analysis. The botto two rows of the table give the mean and standard deviation of these measures across sectors.

TABLE A3
Export patterns in the data

Export outcome	# Obs	Average	St Dev across exporters, importers, and sectors	St Dev of exporter averages	Min	Max
# Trade partners (by exporter-sector)						
Full sample	4347	32.35	41.15	38.05	0	16:
Partners>0	3913	35.94	41.85	37.72	1	16:
Bilateral exports (in logs)	137,490	6.31	2.83	1.15	0	17.7:
Product variety						
SITC-4, full sample	137,490	5.34	6.61	1.97	1	6:
HS-10, exports to U.S.	3933	64.41	147.54	77.39	1	148:

Notes: This table summarizes the variation in export activity across 161 countries and 27 sectors in 1995. A sector

(a) Tables A2-A3: industry characteristics and export patterns.

TABLE A4
Firm selection into exporting vs. firm-level exports

Reported statistic: the contribution of the effect of credit constraints on firm-level exports to the trade-specific effect of credit constraints on bilateral exports

Financial development measure:	Private credit	Repudiation of contracts	Accounting standards	Risk of expropriatio
Panel A. Maximum likelihood estimation				
Fin devt × Ext fin dep (%)	36	63	46	48
Fin devt × Tang (%)	60	78	72	75
Panel B. OLS with polynomial in z_{ijs}				
Fin devt × Ext fin dep (%)	32	61	47	44
Fin devt × Tang (%)	58	77	72	72
Panel C. OLS with 50 bins for predicted probability of exporting				
Fin devt × Ext fin dep (%)	44	68	51	53
Fin devt × Tang (%)	66	81	74	76

(b) Table A4: firm selection vs. firm-level exports.

8 Result map

- **Theory.** Credit constraints raise export cutoffs and reduce export profits, especially in financially vulnerable sectors.
- **Trade versus production.** Finance affects exports beyond its effect on total production.
- **Selection.** Financial development increases the probability that countries export in vulnerable sectors.
- **Product variety.** Financial development expands the number of products shipped abroad.
- **Destinations.** Financial development expands the number of export markets served.
- **Firm-level exports.** Finance also raises average exports conditional on entry.
- **Economic size.** The effects are quantitatively meaningful and comparable to other major sources of trade variation.

9 Short summary

- Financial development is a source of comparative advantage.
- Countries with better finance export relatively more in industries that need more external finance and have less collateral.
- Credit constraints affect trade above and beyond domestic production.
- The effect operates through export participation, product variety, destination coverage, and firm-level exports.

- The results are robust to alternative financial-development measures and controls for factor endowments and institutions.
- The paper links financial development to international trade through heterogeneous-firm selection and export-scale mechanisms.

10 Conclusion

10.1 Core conclusion

Manova shows that credit constraints materially shape international trade. Better financial development gives countries a comparative advantage in financially vulnerable sectors and raises exports through multiple margins.

10.2 Economic intuition

Exporting requires upfront fixed and variable costs. Firms in countries with weak financial systems cannot finance these costs easily, especially in sectors where outside finance needs are high and collateral is scarce. As a result, many productive firms do not export, and exporting firms sell less than they would under better finance.

10.3 Incremental contribution revisited

The paper's incremental contribution is to integrate financial frictions into heterogeneous-firm trade theory and to test the model using country-sector variation in financial vulnerability. It goes beyond showing that finance correlates with trade by decomposing the effect into production, selection, product variety, destinations, and firm-level exports.

10.4 Policy implication

Financial development can promote export diversification and comparative advantage in complex manufacturing sectors. Trade policy alone may not be enough; firms also need access to working capital and financing for fixed export costs.

10.5 Limitations

The design relies on cross-country-sector interactions rather than randomized financial reforms. External finance dependence and tangibility are measured at the sector level and may not capture all firm-level financing needs. The firm-level decomposition is inferred rather than directly observed for all countries.

10.6 Takeaway

Credit constraints are not only a domestic investment friction. They are a trade friction. Financial development determines which firms export, which products are sold abroad, how many destinations are served, and how much exporters sell.